DATA BRIEF · 2025

The Last-Mile Data Gap

Why foundational datasets are the most cost-effective investment in public infrastructure — and why they barely exist.

$15–32	70%	3–5x
ROI per $1 in data	Govts lack infra registries	Planning efficiency gain

The problem

Governments across Sub-Saharan Africa collectively spend over $77 billion per year on infrastructure — energy, water, transport, health, education. Yet the vast majority of these investments are planned without knowing what already exists on the ground.

There is no registry of schools in most countries. No georeferenced inventory of health facilities. No structured dataset of road conditions. No unified view of where electricity reaches and where it does not. The data that would make every infrastructure decision more efficient, more equitable and less wasteful — **simply does not exist**.

This is not a technology problem. Satellite imagery, mobile data collection tools, GIS platforms and cloud infrastructure have existed for over a decade. The gap is institutional, operational and economic: no one has been mandated, funded or equipped to build these foundational datasets at national scale.

The cost of not knowing

The absence of foundational data has measurable consequences. Based on our operational experience across 9 countries and 25+ national-scale projects, we observe four recurring patterns:

— **Duplicated investments.** Without a registry of existing infrastructure, new projects are deployed in areas already served — while underserved zones remain invisible. In one national electrification programme, we found that 12% of planned installations were within 500 metres of existing facilities.

— **Misallocated resources.** Budget allocation based on population estimates rather than actual infrastructure gaps leads to systematic under-investment in rural areas and over-investment in already-served urban zones.

— **Maintenance failures.** Infrastructure that is not inventoried cannot be maintained. Across the countries where we operate, we estimate that 15–25% of public infrastructure is degraded beyond use — but invisible to planning systems.

— **Monitoring blind spots.** Development finance institutions invest billions in infrastructure programmes but lack the ground-truth data to verify impact. Post-project evaluations rely on self-reported data rather than independent spatial evidence.

The economics of foundational data

A 2022 Financial Times analysis estimated that solid information systems generate an average of **$32 in economic returns for every $1 invested in data**. Our own field operations confirm this order of magnitude.

The cost of a national infrastructure census — covering 100% of a country's territory across 20+ sectors — represents a fraction of a single infrastructure deployment project. In Togo, the complete PRISE census (1.25 million assets, 23 sectors, 100% territory coverage) cost less than 0.5% of the annual national infrastructure budget. The geoportal it feeds now serves 23 ministries and informs every major planning decision.

> *"Decision-makers spend more time searching for data — when it exists — than using it. The foundational layer is missing."*

From collection to decision: the full chain

Building foundational datasets is not just about sending field agents with tablets. It requires an integrated chain that connects ground truth to institutional decision-making:

- → **Collect** — Field surveys, remote sensing, census operations. Last-mile coverage with trained local agents at national scale. This is where the data is born — but it is only the beginning.
- → **Structure** — Data engineering, quality assurance, schema design. Raw field data must be cleaned, validated, georeferenced and organised into queryable databases. Without this, collection is wasted effort.
- → **Platform** — Geoportals, dashboards, APIs. The data must be accessible to the institutions that need it — not locked in spreadsheets or consultant reports. Sovereign platforms that governments own and operate.
- → **Analyse** — Spatial analytics, predictive models, deployment simulations. Data becomes intelligence when it answers specific planning questions: where to build, what to prioritise, how to allocate budget.
- → **Govern** — Change management, capacity building, institutional alignment. The most sophisticated platform is useless if ministries do not use it. Sustainable data infrastructure requires organisational transformation.

The organisations that treat data collection as an end in itself — without connecting it to platforms, analytics and governance — perpetuate the gap they claim to address.

What works: lessons from national-scale programmes

Our experience operating in 9 countries suggests that successful foundational data programmes share four characteristics:

- — **Government ownership from day one.** Data sovereignty is not negotiable. The datasets must be owned, hosted and governed by national institutions — not by the consulting firm or donor that funded the collection.
- — **Multi-sector by design.** Single-sector data collection (health only, or energy only) misses the interdependencies that drive planning decisions. A school without electricity. A clinic without a road. These connections only appear in cross-sectoral datasets.
- — **Continuous, not one-off.** A census is not a project — it is an infrastructure. Data must be actualised, maintained and expanded over time. The most effective programmes build permanent institutional capacity for data management.
- — **Connected to decision tools.** Data that sits in a database is not data infrastructure. It becomes infrastructure when it feeds a geoportal that a minister uses to plan, a dashboard that a DFI uses to monitor, an API that a private operator uses to deploy.

About Mitsio Motu — Data & Tech studio. Paris — Lagos — Lomé. We build the data infrastructure that connects ground truth to decision-making. From field census to national geoportal. 25+ projects across 9 countries since 2018.
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